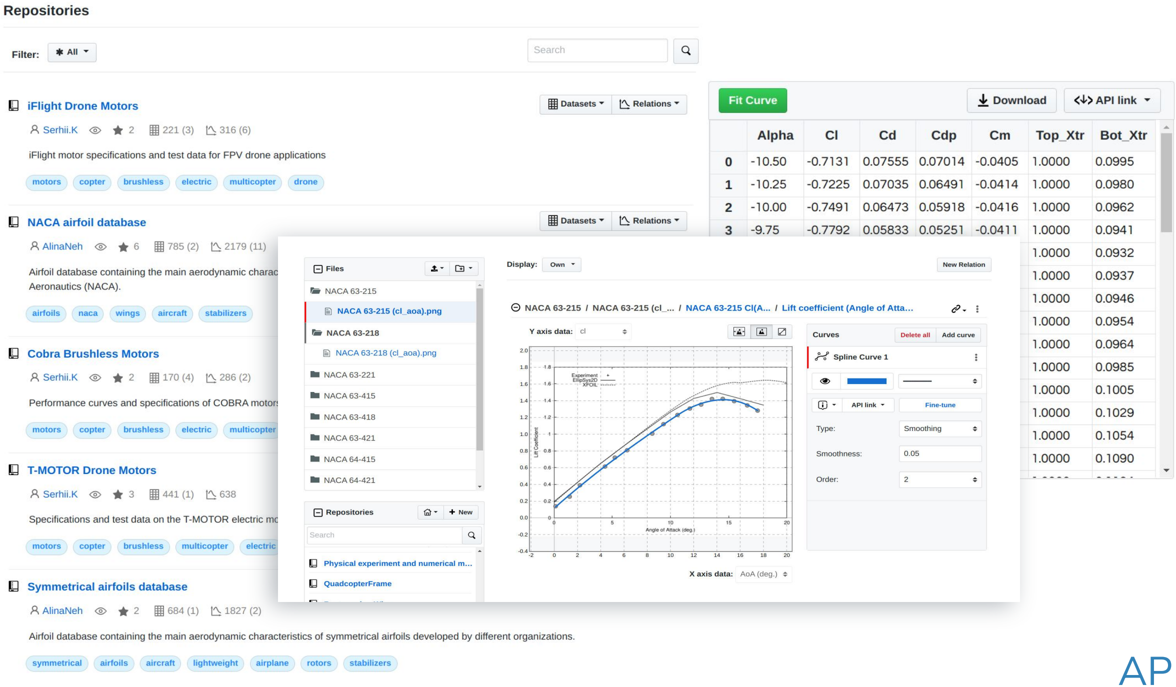
SplineCloud

simplified and reusable curve fitting

SplineCloud : Open Platform for Formalized Knowledge Exchange



Open and free to use 🐧

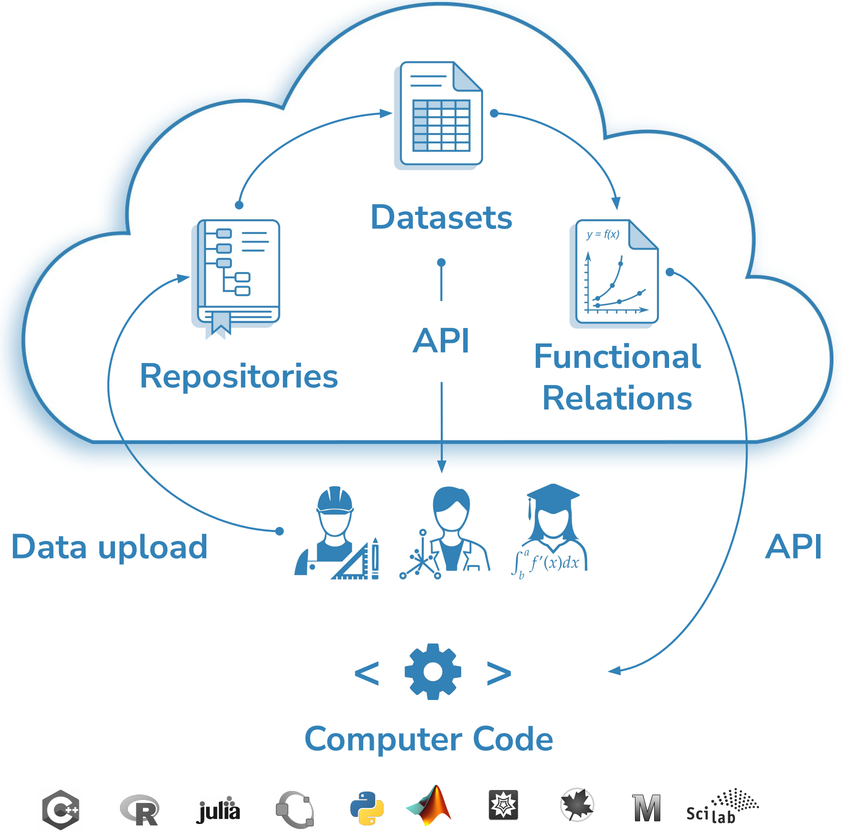
Interactive curve fitting 📉

API for data and functions 💻

SplineCloud

Few Steps to Reusability of Regression Models

1. **Create** topical **repository** (add short description, topics and wiki to make it findable)
2. **Upload data** (spreadsheets, tables, catalogues, graphs)
3. **Define/Create datasets** (use plot digitizing tool to extract data from graphs)
4. **Build** functional **relations** (use advanced spline fitting tool to find relations in data)
5. **Reuse** datasets and curves in code over API



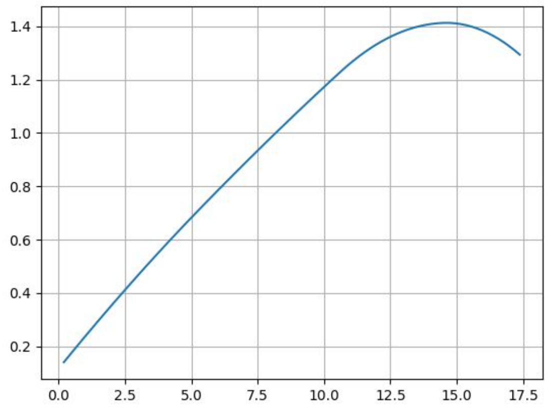
Access and Reuse Spline Curves in Code


```python
from splinecloud_scipy import load_spline

spline = load_spline('spl_K5t56P5bormJ')

X = np.linspace(0, 20, 200)
Y = spline.eval(X)

plt.plot(X, Y)
plt.show()
```

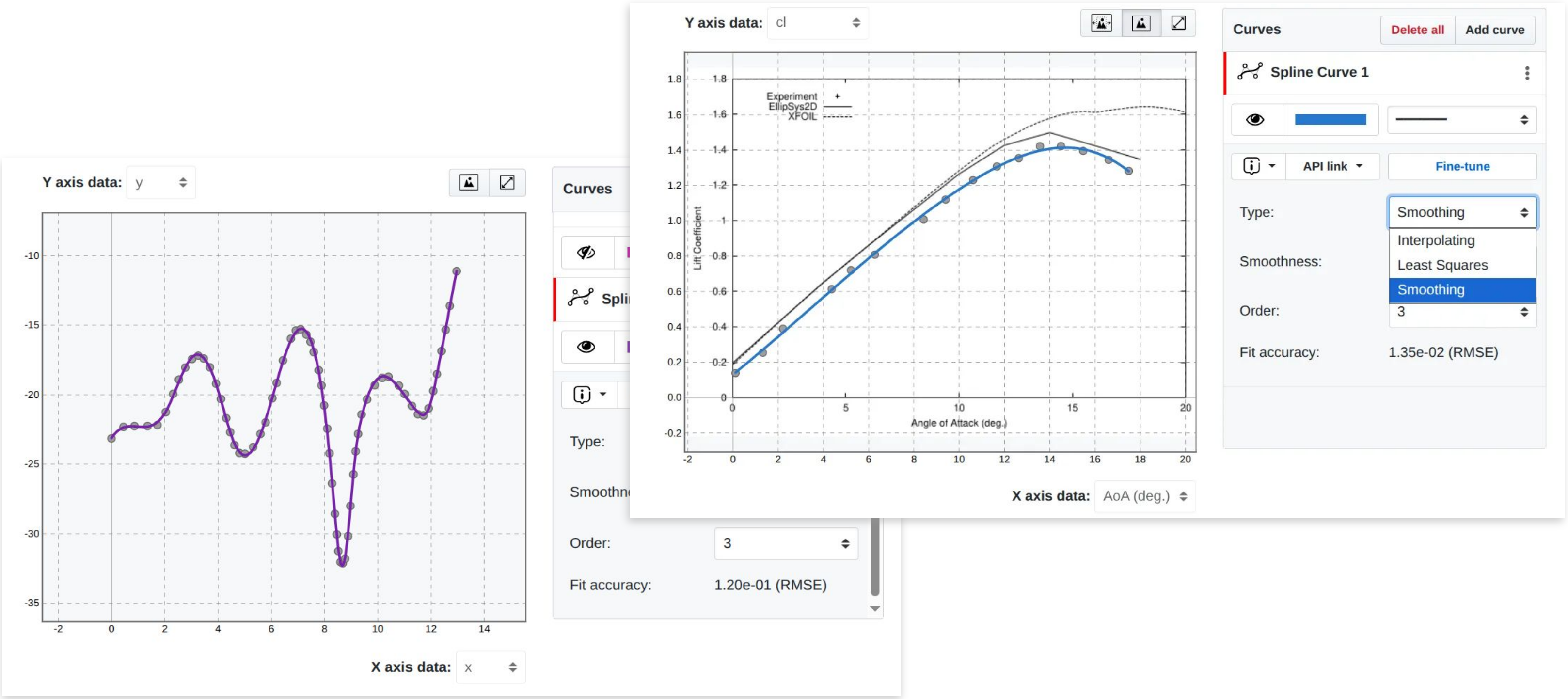


Install client for Python

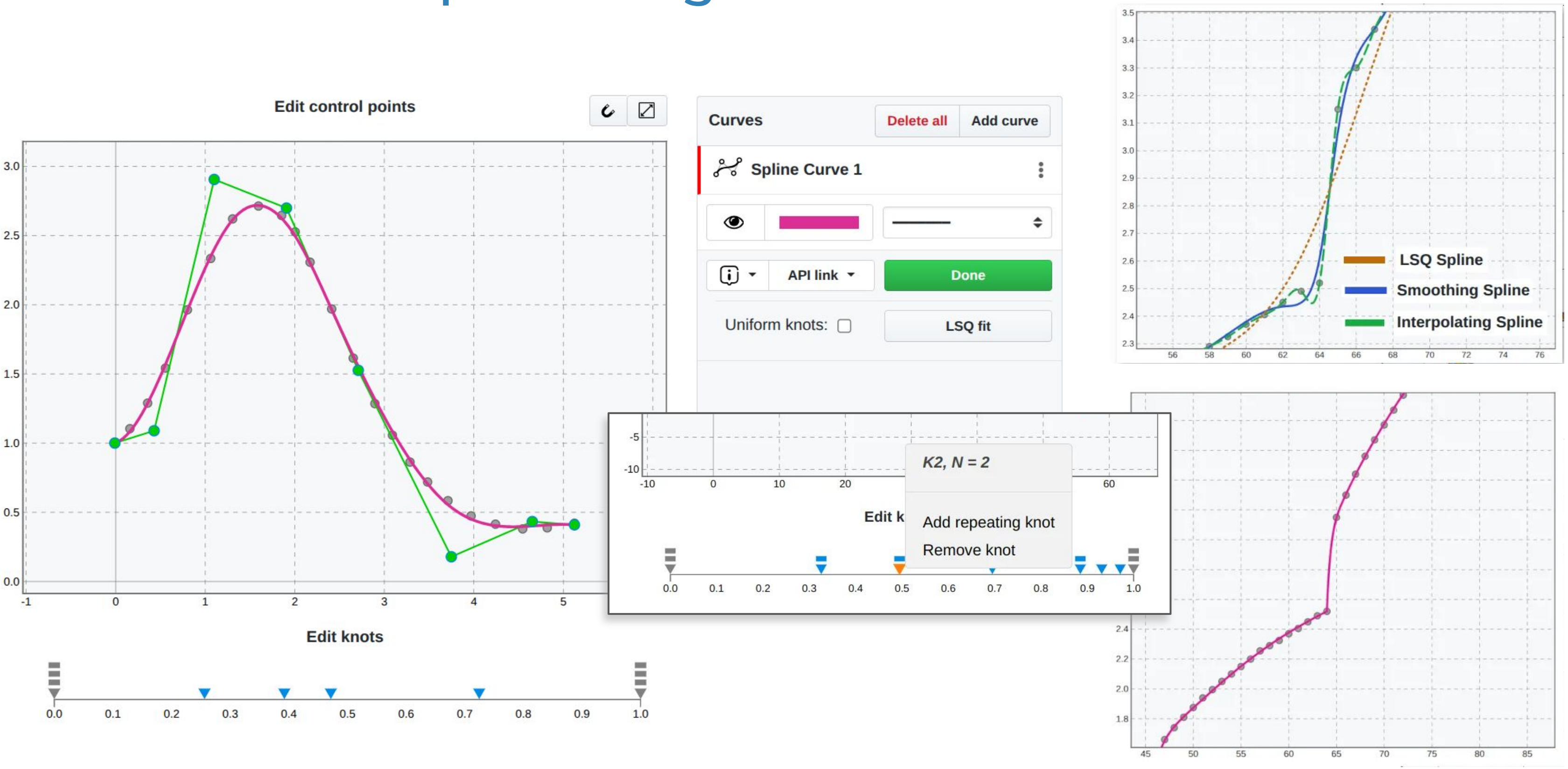
```
pip install splinecloud-scipy
```

Features

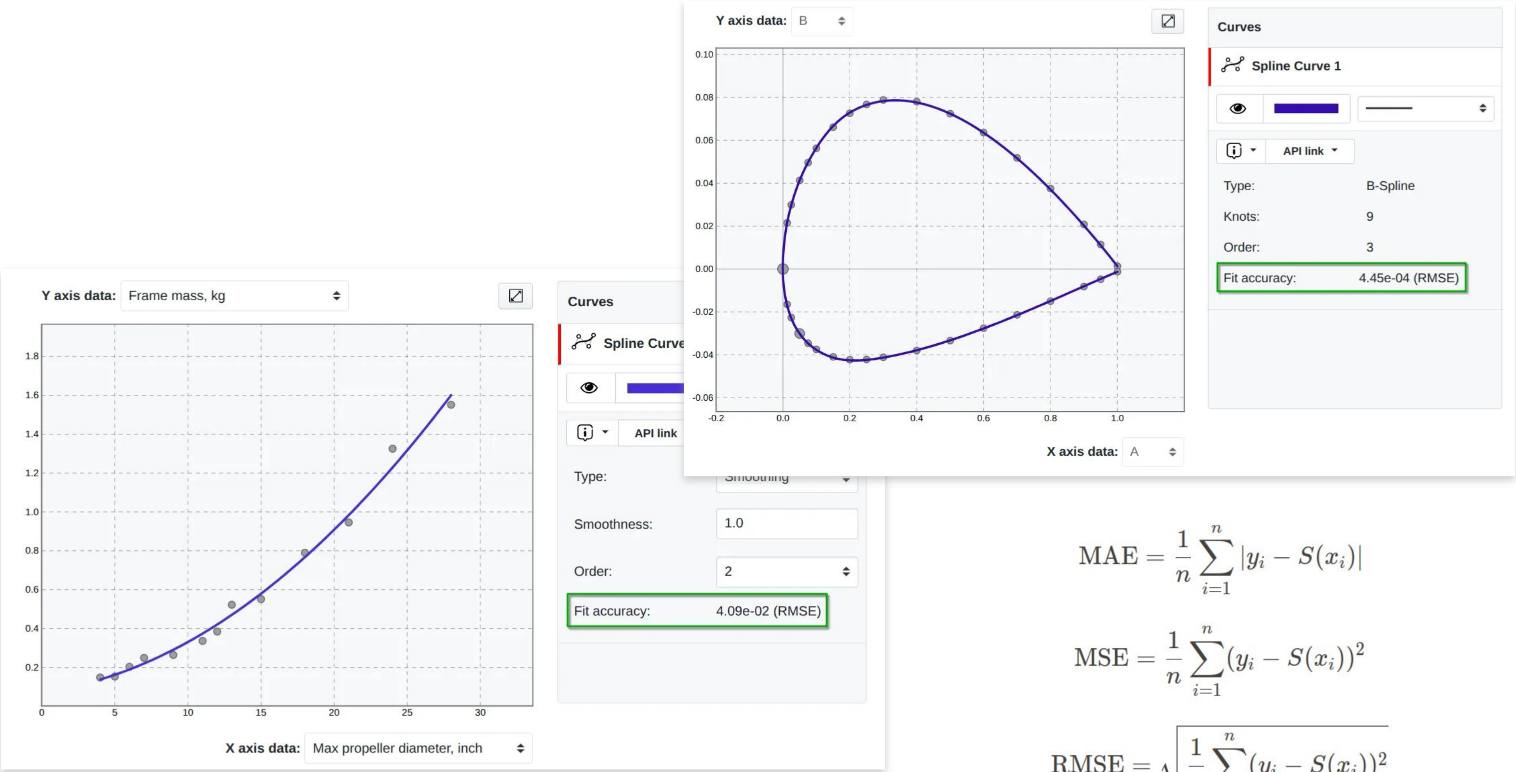
Automatic Spline Fitting Algorithms: SciPy Methods



Interactie Fine Tuning Mode
to Edit Cure Shapes Using Handlers



Estimating Fit Accuracy

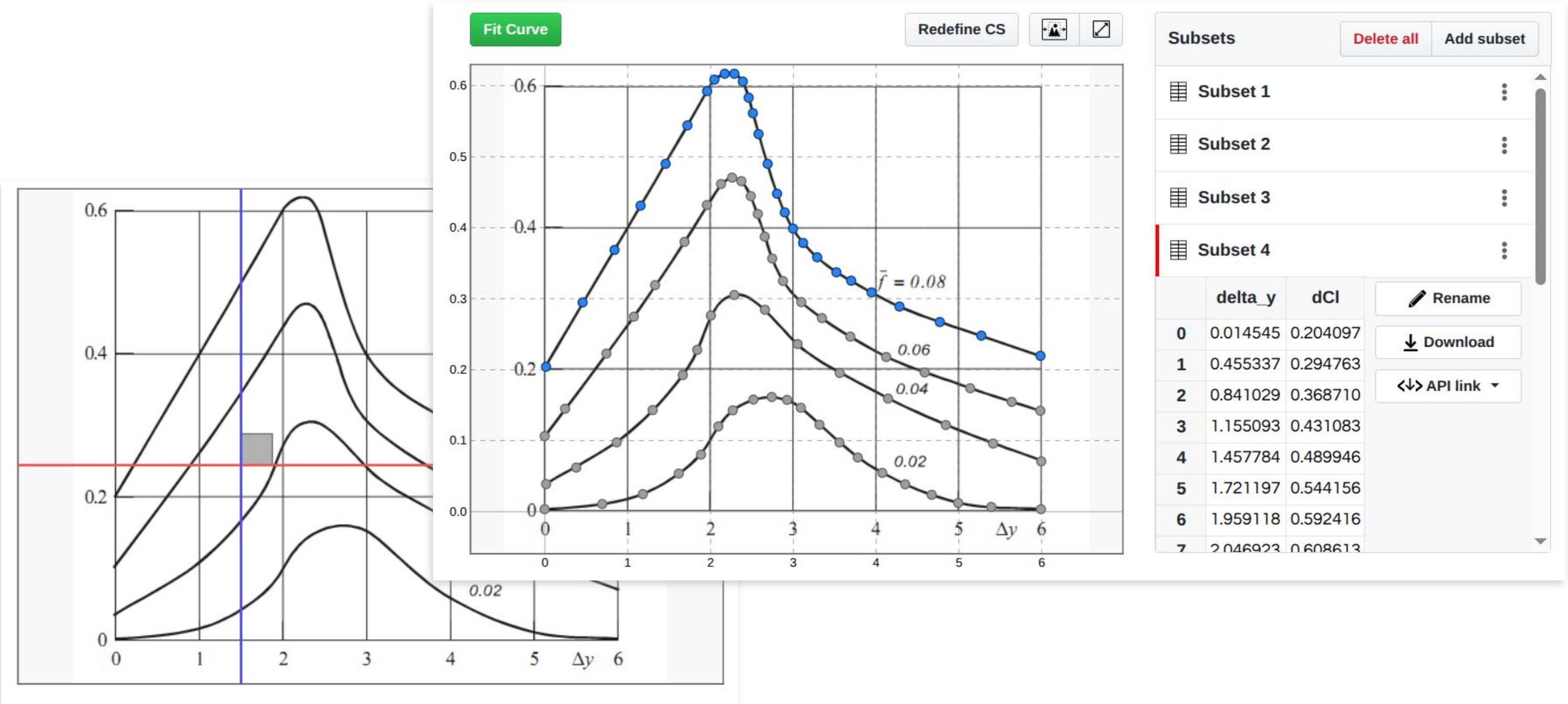


$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - S(x_i)|$$

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} (y_i - S(x_i))^2$$

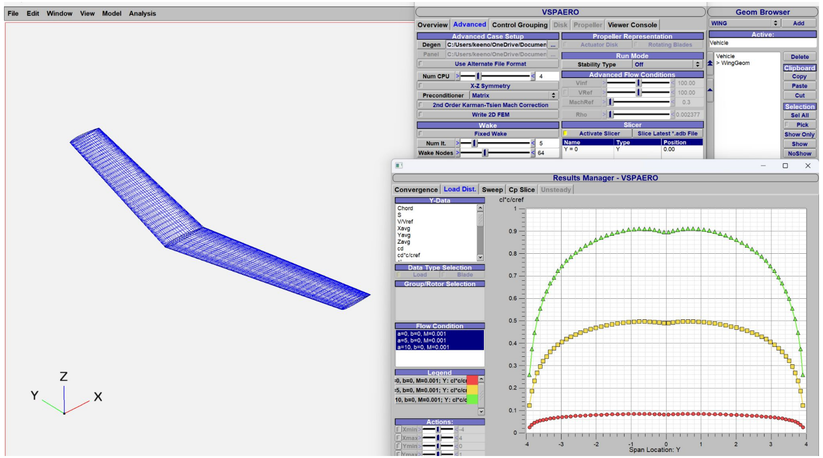
$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - S(x_i))^2}$$

Plot Digitizer

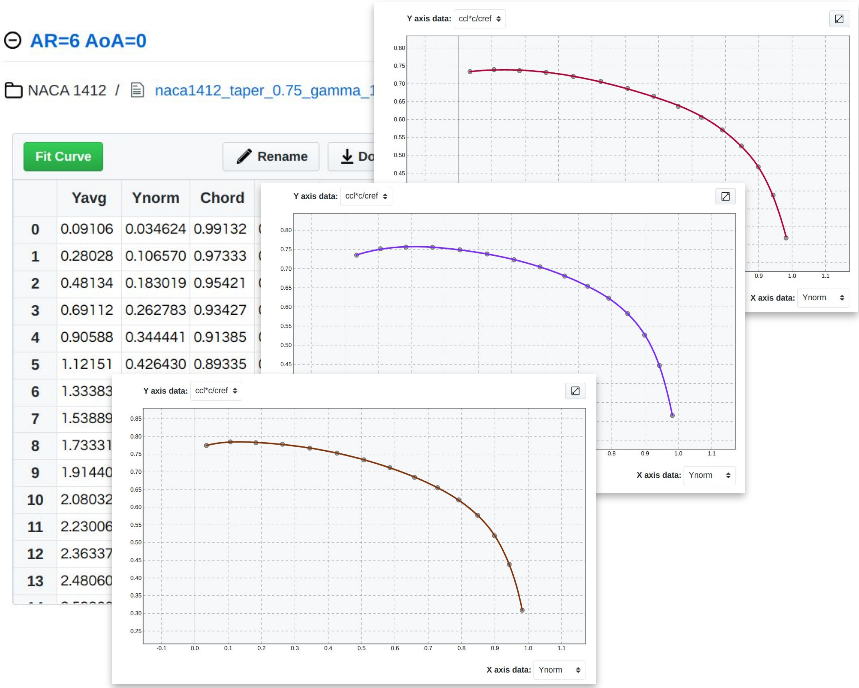


Reusability Examples

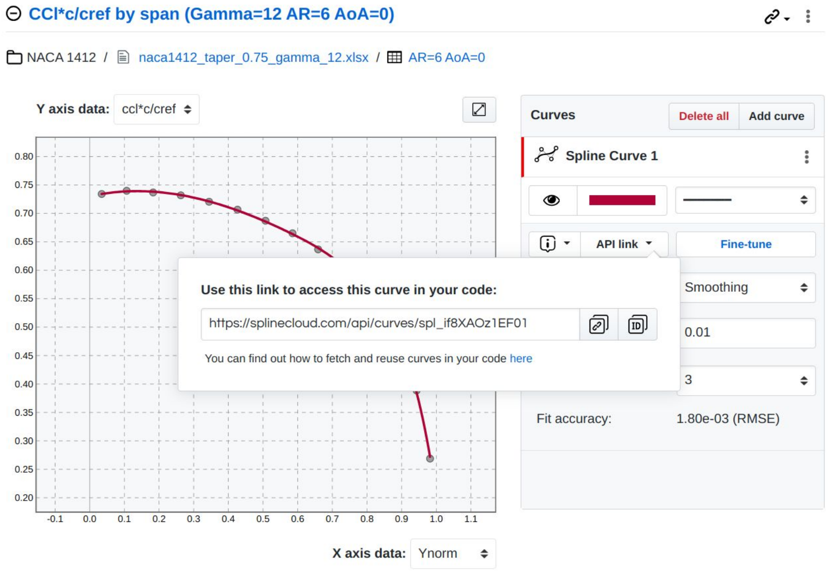
Example: Building Reusable Functional Relations Based on Numerical Analysis



Numerical Analysis in OpenVSP

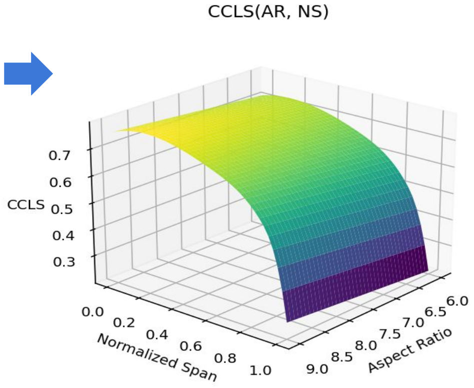


Reusing Collection of Curves in Code




```
ccls_data = {
  "gamma=12": {
        "alpha=0": {
            6: "spl_if8XAOz1EF01",
            7: "spl_QyMqI7K4dyEe",
            8: "spl_qk3ZnLPcdim2",
            9: "spl_rPgCZnOElBc4"
        },
        "alpha=5": {
            6: "spl_JvesBTdOISv5",
            7: "spl_VOSHqg0jfVq1",
            8: "spl_3bGg9cNVK9kj",
            9: "spl_lyEFFznnRiS3"
        },
        "alpha=10": {
            6: "spl_VLW7VrFbfws2",
            7: "spl_aeTTLUK15Byg",
            8: "spl_ORbrtErrHW1h",
            9: "spl_OutDl1pXbXMw"
        }
    },
    "gamma=15": {
        "alpha=0": {
            6: "spl_TDh4VJ5dNXIi",
            7: "spl_SX0MQ3ZNsneq",
            8: "spl_5aTXblNVrGYm",
            9: "spl_Kr8AxTPgoyLd"
        },
        "alpha=5": {
            6: "spl_AtW6LnlClg7z",
            7: "spl_ONuFO7XzDYza",
            8: "spl_BMyNr3WynWlW",
            9: "spl_Tc7Wfoh4tSPb"
        },
        "alpha=10": {
            6: "spl_TJH9zbnkHlSh",
            7: "spl_eu8el2uitQKn",
            8: "spl_2sca03Ca6i1W",
            9: "spl_toFY32FcHMtJ"
        }
    },
```

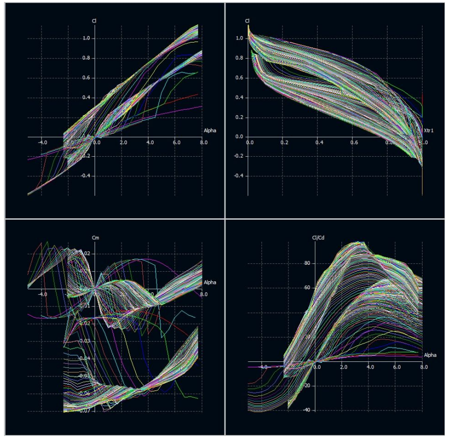
Wing lift efficiency coefficient



https://splinecloud.com/repository/nomad-vagabond/Swept_wing_vorticity_analysis/

Example: Building a Library of Airfoil Performance Curves

XFOIL



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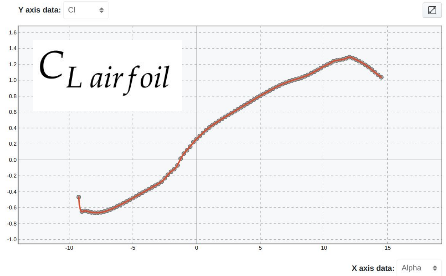

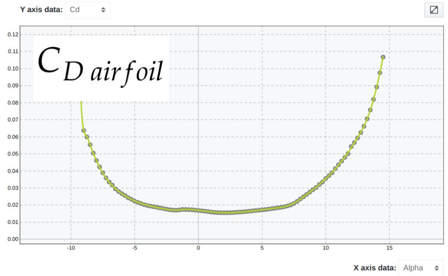

Python


```
{
    "NACA 4412": {
    "name": "NACA 4412 (naca4412-il)",
    "curves": {
        "CL_1000000": "spl_5MJnqWp8tFjk",
        "CL_500000": "spl_cTj0LM7UbRHv",
        "CL_200000": "spl_ajJkUYMsRf2z",
        "CL_100000": "spl_kB9lqsI4roUE",
        "CL_50000": "spl_jlYGh0C0E3HG",
        "CD_1000000": "spl_qYuy0H980Euh",
        "CD_500000": "spl_0tE5L6PgHtLu",
        "CD_200000": "spl_xwV6pCunH6I6",
        "CD_100000": "spl_1hWaK9N0ju3M",
        "CD_50000": "spl_LpjgfXkzt9Xg",
```


```python
class Airfoil(airfoil_data):
    def eval_cl(alpha, Re)
    def eval_cd(alpha, Re)
    def cl_to_cd(alpha, Re)
    def cd_to_cl(alpha, Re)
    def alpha_optimal(Re)
    def alpha_max_lift(Re)
    def alpha_min_drag(Re)
    ...
```

https://splinecloud.com/repository/AlinaNeh/NACA_airfoil_database/

Example: Extracting and Reusing Historical Data

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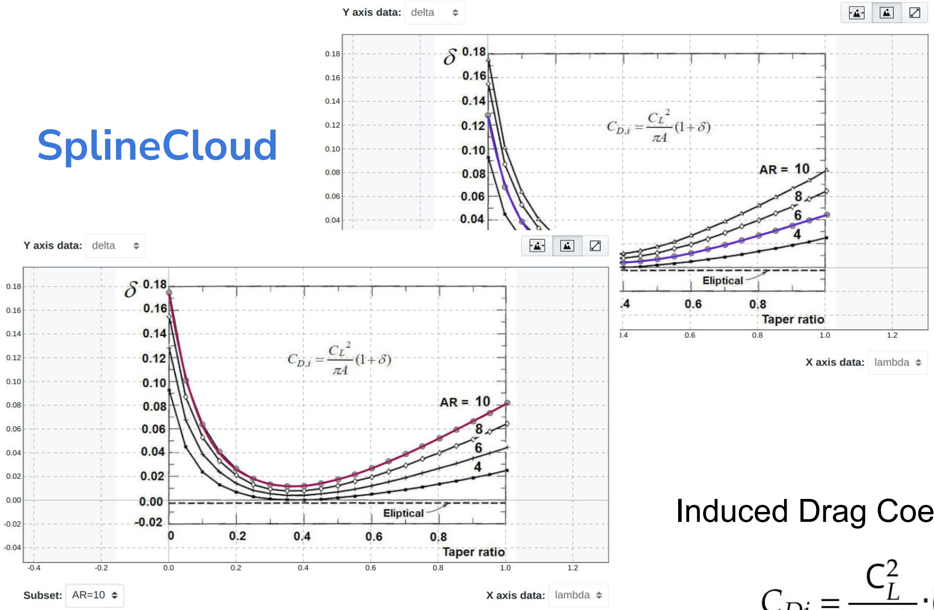
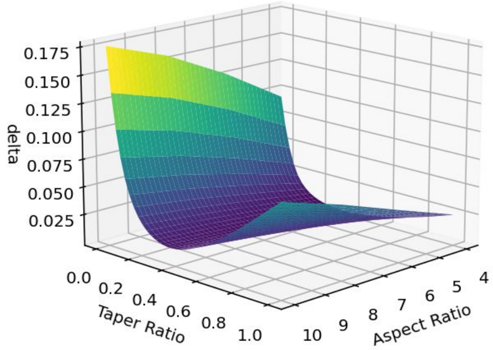


Python

```
delta_curves = {
   "AR=10": load_spline("spl_Nb5m075aHe42"),
   "AR=8":  load_spline("spl_YUtYmXHXXK5Q"),
   "AR=6":  load_spline("spl_VsE31rlqsGtH"),
   "AR=4":  load_spline("spl_lvdZRtwoGU6T")
}
```

Span Eficiency(AR, TR)



Induced Drag Coefficient

$$C_{Di} = \frac{C_L^2}{\pi AR} \cdot (1 + \delta)$$

https://splinecloud.com/compose/dfl_yRvfIsQYc80M/relations

McCormick, B. W. (1979). Aerodynamics, Aeronautics, and Flight Mechanics (p. 652). John Wiley & Sons. https://rexresearch1.com/FlightMechanicsLibrary/AerodynamicAeronauticsFlightMechanics.pdf

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